PART 1: Titanic Dataset – Summary Statistics Grouped by Categorical Variable

In [56]:
```
# Import required libraries
import pandas as pd
import numpy as np
```

In [57]:
```
# Load Titanic dataset
df = pd.read_csv("titanic.csv")
```

In [58]:
```
display(df.head())
```

	PassengerId	Survived	Pclass	Lname	Name	Sex	Age	SibSp	Parch	Ticl
0	1	0	3	Braund	Mr. Owen Harris	male	22.0	1	0	A 211
1	2	1	1	Cumings	Mrs. John Bradley (Florence Briggs Thayer)	female	38.0	1	0	PC 175
2	3	1	3	Heikkinen	Miss. Laina	female	26.0	0	0	STON/(31012
3	4	1	1	Futrelle	Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	1138
4	5	0	3	Allen	Mr. William Henry	male	35.0	0	0	3734

In [59]:
```
# Check column names
print(df.columns)
```

```
Index(['PassengerId', 'Survived', 'Pclass', 'Lname', 'Name', 'Sex', 'Age',
       'SibSp', 'Parch', 'Ticket', 'Fare', 'Cabin', 'Embarked'],
      dtype='object')
```

In [60]:
```
# Select relevant columns (Age = numeric, Pclass = categorical)
data = df[['Age', 'Pclass']]
```

In [61]:
```
# Drop missing values
data = data.dropna() # Removes rows with null values
```

In [62]:
```
# Group data by categorical variable 'Pclass'
grouped = data.groupby('Pclass')   # Groups passengers by class (1,2,3)
```

In [63]:
```
# Calculate summary statistics
summary = grouped['Age'].agg(['mean', 'median', 'min', 'max', 'std'])
```

In [64]:
```
# Display summary statistics
display(summary)   # Shows mean, median, min, max, std for each class
```

Pclass	mean	median	min	max	std
1	38.111111	38.0	19.00	71.0	14.744838
2	28.114828	29.0	0.83	66.0	14.342216
3	24.307143	22.0	2.00	70.5	12.949538

In [65]:
```
# Create list of Age values for each Pclass
age_list = grouped['Age'].apply(list)
```

In [66]:
```
# Display list
print(age_list)   # Shows list of ages for each passenger class
```

```
Pclass
1    [38.0, 35.0, 54.0, 58.0, 28.0, 19.0, 40.0, 28....
2    [14.0, 55.0, 35.0, 34.0, 66.0, 27.0, 3.0, 29.0...
3    [22.0, 26.0, 35.0, 2.0, 27.0, 4.0, 20.0, 39.0,...
Name: Age, dtype: object
```

PART 2: Iris Dataset – Statistical Details by Species

In [67]:
```
# Load Iris dataset
df = pd.read_csv("iris.csv")
```

In [68]:
```
# Display first 5 rows
display(df.head())
```

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa

In [69]:
```
# Check unique species
print(df['species'].unique())   # Displays different species
```

```
['setosa' 'versicolor' 'virginica']
```

In [70]:
```
# Group dataset by species
grouped = df.groupby('species')   # Groups data by species
```

In [71]:
```
# Calculate mean for each species
display(grouped.mean())   # Mean values of each feature per species
```

species	sepal_length	sepal_width	petal_length	petal_width
setosa	5.006	3.418	1.464	0.244
versicolor	5.936	2.770	4.260	1.326
virginica	6.588	2.974	5.552	2.026

In [72]:
```
# Calculate standard deviation
display(grouped.std())   # Standard deviation for each species
```

species	sepal_length	sepal_width	petal_length	petal_width
setosa	0.352490	0.381024	0.173511	0.107210
versicolor	0.516171	0.313798	0.469911	0.197753
virginica	0.635880	0.322497	0.551895	0.274650

In [73]:
```
# Calculate percentiles (25%, 50%, 75%)
display(grouped.quantile([0.25, 0.5, 0.75]))   # Percentile values
```

species		sepal_length	sepal_width	petal_length	petal_width
setosa	**0.25**	4.800	3.125	1.400	0.2
	0.50	5.000	3.400	1.500	0.2
	0.75	5.200	3.675	1.575	0.3
versicolor	**0.25**	5.600	2.525	4.000	1.2
	0.50	5.900	2.800	4.350	1.3
	0.75	6.300	3.000	4.600	1.5
virginica	**0.25**	6.225	2.800	5.100	1.8
	0.50	6.500	3.000	5.550	2.0
	0.75	6.900	3.175	5.875	2.3

In [74]:
```
# Describe full statistics
display(grouped.describe())   # Complete statistical summary
```

	sepal_length								sepal_width		...	petal_
	count	mean	std	min	25%	50%	75%	max	count	mean	...	75%
species												
setosa	50.0	5.006	0.352490	4.3	4.800	5.0	5.2	5.8	50.0	3.418	...	1.575
versicolor	50.0	5.936	0.516171	4.9	5.600	5.9	6.3	7.0	50.0	2.770	...	4.600
virginica	50.0	6.588	0.635880	4.9	6.225	6.5	6.9	7.9	50.0	2.974	...	5.875

3 rows × 32 columns